Unit-3: Introduction to Probability

Topic 1: Introduction to Probability Theory

Probability is one of the **most important mathematical foundations of Data Science and Machine Learning**. Many algorithms (Naïve Bayes, Logistic Regression, Bayesian models, etc.) are built on probability concepts.

1. What is Probability

Definition: Probability is the measure of the likelihood that an event will occur.

It tells us **how likely something is to happen**.

Probability values always lie between:

0 ≤ Probability ≤ 1

Probability Value	Meaning
0	impossible event
1	certain event
between 0 and 1	possible event

2. Basic Example

Coin Toss: When we toss a coin, two outcomes are possible:

- Head (H)
- Tail (T)

Probability of getting Head:

P(H) = 1 / 2

Probability of getting Tail:

P(T) = 1 / 2

Both outcomes are **equally likely**.

3. Probability Formula

The classical probability formula is:

P(E) = Number of favorable outcomes / Total number of possible outcomes

Where:

Symbol	Meaning
P(E)	probability of event E
Favorable outcomes	outcomes where event occurs
Total outcomes	all possible outcomes

4. Example: Rolling a Dice

A standard dice has **6 faces**.

Possible outcomes:

1, 2, 3, 4, 5, 6

Total outcomes = **6**

Example 1: Probability of getting 3

Favorable outcomes = {3}

P(3) = 1 / 6

Example 2: Probability of getting an even number

Even numbers = {2, 4, 6}

Favorable outcomes = **3**

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P(Even) = 3 / 6 = 1 / 2

5. Key Terms in Probability
Understanding probability requires several basic terms.

Term	Meaning
Experiment	process that produces outcomes
Outcome	result of experiment
Event	set of outcomes
Sample Space	set of all possible outcomes

6. Random Experiment
Definition: A **random experiment** is an experiment whose outcome cannot be predicted with certainty.
Examples:

Experiment	Possible Outcomes
coin toss	head or tail
dice roll	numbers 1–6
card draw	52 possible cards

7. Outcome
Definition: An **outcome** is a possible result of a random experiment.
Example:
Experiment: Roll a dice
Outcomes:
1, 2, 3, 4, 5, 6
Each number is an **outcome**.

8. Sample Space
Definition: The **sample space** is the set of all possible outcomes of an experiment.
Symbol: S

Example 1: Coin Toss
S = {H, T}

Example 2: Dice Roll
S = {1,2,3,4,5,6}

9. Event
Definition: An **event** is a subset of the sample space.
Example:
Dice experiment
S = {1,2,3,4,5,6}
Event A = Getting an even number
A = {2,4,6}
Event B = Getting number greater than 4
B = {5,6}

10. Types of Events
Simple Event: An event with **only one outcome**.

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Example: Getting 4 when rolling a dice

Compound Event: an event with **more than one outcome**.
Example: Getting even number {2,4,6}

11. Complementary Event
If event **A** occurs, event **A'** (complement) does not occur.
Example:
Coin toss
A = Head
A' = Tail
Relationship:
P(A) + P(A') = 1

12. Real-World Examples of Probability
Probability is used in many fields.

Field	Application
Weather Forecasting	probability of rain
Finance	stock market risk
Healthcare	disease prediction
Machine Learning	classification models
Gaming	predicting outcomes

Example in Data Science
Spam detection systems calculate:
P(email is spam | contains certain words)
This is the principle behind **Bayesian classifiers**.

13. Importance in Data Science
Probability helps in:

Area	Purpose
uncertainty modeling	handle randomness
predictive models	estimate outcomes
statistical inference	analyze data patterns
machine learning	build probabilistic models

Many ML algorithms rely on probability distributions.

Final Concept Summary

Concept	Meaning
Probability	likelihood of event
Sample Space	set of all outcomes
Event	subset of outcomes
Experiment	process producing outcomes

Core Idea
Random Experiment
↓
Sample Space
↓

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Events
↓
Probability of Events

Topic 2: Probability Theory – Terminology

To understand probability properly, several **basic terms and concepts** must be clearly defined. These terms form the foundation for studying **probability distributions, machine learning models, and statistical inference**.

1. Random Experiment

Definition: A **random experiment** is a process or action whose outcome cannot be predicted with certainty in advance, even though all possible outcomes are known.

Examples

Experiment	Possible Outcomes
Tossing a coin	Head or Tail
Rolling a dice	1, 2, 3, 4, 5, 6
Drawing a card from deck	52 possible cards
Weather tomorrow	rain, no rain

The key property is **uncertainty of the result**.

6. Certain Event

Definition: A **certain event** is an event that will always occur.
Probability:
$P(E) = 1$
Example:
Rolling a dice → getting number ≤ 6
$E = \{1,2,3,4,5,6\}$

7. Impossible Event

Definition : An **impossible event** cannot occur.
Probability:
$P(E) = 0$
Example:
Rolling a dice → getting 7

8. Mutually Exclusive Events

Definition: Two events are **mutually exclusive** if they cannot occur at the same time.

Example

Dice experiment
Event A = getting 2
Event B = getting 5
Both cannot occur simultaneously.

Mathematical Representation

$A \cap B = \varnothing$
Example:

Event	Outcome

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A	{2}
B	{5}

No common elements.

9. Non-Mutually Exclusive Events
Events that **can occur together**.

Example
Dice experiment
Event A = even number
A = {2,4,6}
Event B = number greater than 3
B = {4,5,6}
Common outcomes:
{4,6}
So these events are **not mutually exclusive**.

10. Independent Events
Definition: Two events are **independent** if the occurrence of one event does not affect the probability of the other event.

Example
Experiment:

- Toss a coin
- Roll a dice

Outcome of coin toss **does not affect** dice result.
Therefore, events are **independent**.

Formula
$P(A \cap B) = P(A) \times P(B)$

11. Dependent Events
Definition : Events are **dependent** if the occurrence of one event affects the probability of another event.

Example
Drawing two cards from a deck **without replacement**.
First card affects probability of second card.
Example:
Probability of drawing an Ace changes after one card is removed.

12. Complementary Event
Definition : The **complement of an event A** includes all outcomes where event A does not occur.
Symbol: A'

Example
Coin Toss
Event A = Head
A = {H}

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Complement:
A' = {T}

Formula
P(A) + P(A') = 1

13. Union of Events
Union represents **either event occurring**.
Symbol:
A ∪ B
Meaning:
- event A occurs
- OR event B occurs
- OR both occur

Example
Dice experiment
Event A = even numbers
{2,4,6}
Event B = numbers greater than 4
{5,6}
Union:
{2,4,5,6}

14. Intersection of Events
Intersection represents **both events occurring together**.
Symbol:
A ∩ B
Example:
A = {2,4,6}
B = {4,5,6}
Intersection:
{4,6}

Summary of Important Probability Terms

Term	Meaning
Random Experiment	process with uncertain outcome
Outcome	result of experiment
Sample Space	set of all outcomes
Event	subset of sample space
Mutually Exclusive	cannot occur together
Independent Events	occurrence of one does not affect another
Complementary Event	opposite of event
Union	A or B
Intersection	A and B

Concept Map
Random Experiment
↓

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Sample Space
↓
Events
↓
Event Relationships
(Union, Intersection, Complement, Independence)

Terminology and Fundamental Concepts – Axioms of Probability

Probability theory is built on a few **basic concepts and rules (axioms)** that define how probabilities behave. These rules ensure that probability calculations are **consistent and mathematically valid**.

1. Basic Terminology in Probability

Understanding probability begins with some essential terms.

Random Experiment

A **random experiment** is a process that produces an outcome that cannot be predicted with certainty.

Examples:

Experiment	Possible Outcomes
Toss a coin	Head or Tail
Roll a dice	1,2,3,4,5,6
Draw a card	52 possible cards

Outcome: An **outcome** is a single possible result of a random experiment.

Example:

Rolling a dice → outcomes = {1,2,3,4,5,6}

Each number is an outcome.

3. Event Relationships

Mutually Exclusive Events

Two events are **mutually exclusive** if they cannot occur simultaneously.

Example:

Dice experiment

Event A = getting 2

Event B = getting 5

Both cannot occur at the same time.

Mathematically:

$A \cap B = \varnothing$

Independent Events

Two events are **independent** if the occurrence of one event does not affect the probability of the other event.

Example:

- Tossing a coin
- Rolling a dice

These two outcomes do not affect each other.

Formula:

$P(A \cap B) = P(A) \times P(B)$

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Complementary Event
If event **A** occurs, event **A′** does not occur.
Example:
Coin toss
A = Head
A′ = Tail
Formula:
$P(A) + P(A') = 1$

4. Fundamental Concepts of Probability
The modern mathematical theory of probability is based on **axioms introduced by the mathematician Andrey Kolmogorov**.
These axioms define the **rules that probability values must follow**.

5. Axioms of Probability
There are **three basic axioms of probability**.

Axiom 1: Non-Negativity
The probability of any event is always **non-negative**.
$P(A) \geq 0$
Meaning:
Probability cannot be negative.
Example:
P(rain tomorrow) ≥ 0

Axiom 2: Total Probability of Sample Space
The probability of the **entire sample space is equal to 1**.
$P(S) = 1$
Meaning:
One of the outcomes in the sample space **must occur**.
Example:
Dice experiment:
$S = \{1,2,3,4,5,6\}$
Total probability of all outcomes = 1.

Axiom 3: Additivity Rule
If two events are **mutually exclusive**, the probability that either event occurs is the sum of their probabilities.
$P(A \cup B) = P(A) + P(B)$
Condition:
$A \cap B = \varnothing$

Example
Dice experiment:
Event A = getting 2
$P(A) = 1/6$
Event B = getting 5
$P(B) = 1/6$
Both cannot occur together.

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So:
$P(A \cup B) = 1/6 + 1/6 = 2/6$

6. Additional Probability Rules
Using axioms, several useful formulas can be derived.

Complement Rule
$P(A') = 1 - P(A)$
Example:
Probability of rain = 0.7
Probability of no rain:
$1 - 0.7 = 0.3$

General Addition Rule
For events that are **not mutually exclusive**:
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
This avoids counting overlapping outcomes twice.

7. Example Using Addition Rule
Dice experiment:
Event A = even number
$A = \{2,4,6\}$
Event B = number greater than 3
$B = \{4,5,6\}$
Intersection:
$A \cap B = \{4,6\}$
Probability:
$P(A) = 3/6$
$P(B) = 3/6$
$P(A \cap B) = 2/6$
Using formula:
$P(A \cup B) = 3/6 + 3/6 - 2/6$
$= 4/6$

8. Importance in Data Science
Probability axioms are used in:

Area	Application
Machine Learning	Bayesian models
Statistics	hypothesis testing
AI systems	predictive modeling
Risk analysis	uncertainty estimation

Many algorithms rely on probability rules.
Example:
Naïve Bayes classifier calculates:
$P(\text{Class} \mid \text{Data})$
using probability theory.

Probability Structure

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Random Experiment
↓
Sample Space
↓
Events
↓
Probability Rules
(Axioms)

Conditional Probability

Conditional probability is one of the **most important concepts in probability and machine learning**.
It measures the probability of an event **given that another event has already occurred**.
Many real-world systems such as **spam filters, medical diagnosis, and recommendation systems** use conditional probability.

1. Definition of Conditional Probability
Conditional probability is the probability of an event occurring given that another event has already occurred.
It answers the question:
"What is the probability of event A occurring if event B has already happened?"
Notation:
P(A | B)
Meaning:

Symbol	Meaning
P(A	B)

2. Conditional Probability Formula
The mathematical formula is:
$P(A \mid B) = P(A \cap B) / P(B)$
Where:

Term	Meaning
$P(A \cap B)$	probability that both events occur
P(B)	probability that event B occurs

Condition:
$P(B) \neq 0$

3. Intuition Behind Conditional Probability
When event **B has already occurred**, the sample space becomes **smaller**.
So we only consider **outcomes where B happens**.
Then we calculate how many of those outcomes also satisfy **event A**.

4. Example 1 – Dice Experiment
Experiment: Roll a dice
Sample space:
S = {1,2,3,4,5,6}
Event A = even number
A = {2,4,6}
Event B = number greater than 3

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$B = \{4,5,6\}$

Step 1: Find intersection
$A \cap B = \{4,6\}$
Probability:
$P(A \cap B) = 2/6$

Step 2: Find probability of B
$P(B) = 3/6$

Step 3: Apply formula
$P(A \mid B) = (2/6) / (3/6)$
$= 2/3$
Meaning:
If we know the number is **greater than 3**, the probability that it is **even** is **2/3**.

5. Example 2 – Card Problem
A standard deck has **52 cards**.
Suppose:
Event A = drawing a **King**
Event B = drawing a **Face card**
Face cards =
J, Q, K
Total face cards = **12**
Kings = **4**

Conditional probability
$P(\text{King} \mid \text{Face Card})$
Formula:
$P(\text{King} \mid \text{Face Card}) = 4 / 12$
$= 1 / 3$
Meaning:
If we know the card is a **face card**, the probability it is a **king** is **1/3**.

6. Conditional Probability and Independence
Two events are **independent** if:
$P(A \mid B) = P(A)$
Meaning:
The occurrence of event **B does not affect A**.
Example:

- Tossing a coin
- Rolling a dice

These two experiments are independent.

7. Multiplication Rule
From conditional probability we derive the **multiplication rule**.
$P(A \cap B) = P(A \mid B) \times P(B)$
or
$P(A \cap B) = P(B \mid A) \times P(A)$

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This rule helps calculate the probability of **two events occurring together**.

8. Example Using Multiplication Rule
Suppose:
$P(A \mid B) = 0.4$
$P(B) = 0.5$
Then:
$P(A \cap B) = 0.4 \times 0.5$
$= 0.2$

9. Real-World Applications
Conditional probability is widely used in many fields.

Field	Application
Healthcare	disease probability given symptoms
Email systems	spam detection
Finance	risk assessment
Weather forecasting	probability of rain given cloud conditions
Machine learning	Bayesian models

Example in Data Science
Spam filtering uses conditional probability:
P(Spam | Email Content)
Meaning: Probability that an email is spam **given certain words appear in the email**.
This is the principle behind **Naïve Bayes classifiers**.

10. Relationship with Bayes' Theorem
Conditional probability leads to **Bayes' theorem**, which is widely used in machine learning.
Bayes formula:
$P(A \mid B) = [P(B \mid A) \times P(A)] / P(B)$
This formula helps update probabilities when **new information is available**.

Key Idea
New Information (B)
↓
Reduced Sample Space
↓
Updated Probability of A
Conditional probability allows us to **update probabilities when additional information becomes available**.

Bayes' Theorem

Bayes' Theorem is a very important concept in probability and data science. It helps update the **probability of an event when new information becomes available**.
Many machine learning algorithms such as **Naïve Bayes classifiers** are based on this theorem.

1. Definition of Bayes' Theorem
Bayes' Theorem is used to calculate the conditional probability of an event based on prior knowledge of related events.

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It allows us to reverse conditional probabilities.
Example idea:
Instead of calculating
P(Evidence | Hypothesis)
we calculate
P(Hypothesis | Evidence)

2. Bayes' Theorem Formula
The mathematical formula is:
P(A | B) = [P(B | A) × P(A)] / P(B)

3. Components of Bayes' Theorem
1. Prior Probability
The **prior probability** is the probability of an event before observing new data.
Example:
P(Disease)
Probability that a person has a disease before seeing test results.

2. Likelihood
Likelihood represents the probability of observing the evidence given the hypothesis.
Example:
P(Test positive | Disease)

3. Evidence
Evidence is the total probability of observing the data.
Example:
P(Test positive)

4. Posterior Probability
Posterior probability is the **updated probability after observing evidence**.
Example:
P(Disease | Test positive)

4. Example of Bayes' Theorem
Suppose:
- 1% of people have a disease
- Test accuracy for detecting disease = 99%

Let:
P(Disease) = 0.01
P(Positive | Disease) = 0.99
P(Positive | No Disease) = 0.05
We want:
P(Disease | Positive)
Using Bayes' theorem:
P(Disease | Positive) =
[P(Positive | Disease) × P(Disease)] / P(Positive)
First compute P(Positive):

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P(Positive) =
P(Positive | Disease)P(Disease)
+ P(Positive | No Disease)P(No Disease)
P(Positive) =
(0.99 × 0.01) + (0.05 × 0.99)
P(Positive) = 0.0099 + 0.0495 = 0.0594
Now apply Bayes formula:
P(Disease | Positive) =
0.0099 / 0.0594 ≈ 0.1667
So the probability is about **16.67%**.
This example shows how **probabilities change when new information is available**.

5. Intuition Behind Bayes' Theorem
Bayes' theorem works like this:
Prior Knowledge
↓
Observe Evidence
↓
Update Probability
↓
Posterior Probability
This process is called **Bayesian inference**.

6. Example: Spam Email Detection
Suppose we want to calculate:
P(Spam | Word = "Free")
Using Bayes theorem:
P(Spam | Free) =
[P(Free | Spam) × P(Spam)] / P(Free)
This is exactly how **Naïve Bayes spam filters work**.

7. Applications of Bayes' Theorem

Field	Application
Machine Learning	Naïve Bayes classifier
Medical diagnosis	disease prediction
Finance	risk analysis
Email systems	spam detection
AI systems	probabilistic reasoning

8. Importance in Data Science
Bayes' theorem is important because it allows:
- updating probabilities with new data
- building probabilistic models
- making predictions under uncertainty

Many machine learning models use **Bayesian reasoning**.

Final Summary

Concept	Meaning

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Prior	initial probability
Likelihood	probability of evidence
Evidence	probability of observed data
Posterior	updated probability

Bayes' Theorem Structure
Posterior = (Likelihood × Prior) / Evidence

Random Variables

Random variables are the **bridge between probability theory and numerical data**. They convert outcomes of random experiments into **numerical values that can be analyzed mathematically**.
This concept is extremely important because **probability distributions and machine learning models use random variables**.

1. Definition of Random Variable
A **random variable** is a variable whose value is determined by the outcome of a random experiment.
In simple terms:
Random Variable = Numerical representation of outcomes of a random experiment
Symbol commonly used: X

2. Example: Coin Toss
Experiment: Toss a coin
Sample space:
S = {Head, Tail}
Define random variable **X** as:

Outcome	Random Variable X
Head	1
Tail	0

Here the outcomes are converted into **numbers**.

3. Example: Rolling a Dice
Experiment: Roll a dice
Sample space:
S = {1,2,3,4,5,6}
Random variable **X** = number obtained on dice.

Outcome	X
1	1
2	2
3	3
4	4
5	5
6	6

Here the random variable directly represents the outcome.

4. Why Random Variables Are Important
Random variables allow us to:

- apply **mathematical analysis**
- calculate **expected values**

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- describe **probability distributions**
- build **statistical and machine learning models**

Without random variables, probability would only describe **events**, not **numerical data**.

5. Types of Random Variables

There are **two main types of random variables**.

Type	Description
Discrete Random Variable	takes countable values
Continuous Random Variable	takes infinite values in a range

6. Discrete Random Variable

Definition : A **discrete random variable** takes **countable distinct values**.
These values are usually **integers**.

Examples

Experiment	Random Variable
number of heads in 3 coin tosses	0,1,2,3
number of customers entering a store	0,1,2,3...
number of defective products	0,1,2...

Example: Tossing 2 Coins

Sample space:
S = {HH, HT, TH, TT}
Let **X = number of heads**

Outcome	X
HH	2
HT	1
TH	1
TT	0

Possible values of X:
{0,1,2}
This is a **discrete random variable**.

7. Continuous Random Variable

Definition: A **continuous random variable** can take **any value within a range**.
Values are obtained through **measurement**.

Examples

Variable	Example Values
height of a person	170.2 cm
temperature	36.5°C
time taken to complete task	4.35 seconds
weight of object	2.67 kg

These values can include **decimal numbers**.

Example

Let **X = height of students**
Possible values:

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150.2 cm
165.7 cm
178.3 cm
There are **infinitely many possible values**.

8. Difference Between Discrete and Continuous Random Variables

Feature	Discrete Random Variable	Continuous Random Variable
Values	countable	infinite
Type	integers	real numbers
Example	number of students	height, weight
Distribution	PMF	PDF

9. Representation of Random Variables
A random variable is often represented as:
$X : S \rightarrow R$
Meaning:
Random variable **X** maps outcomes from the sample space **S** to real numbers **R**.

10. Real-World Applications
Random variables are used in many areas.

Field	Example
Finance	stock price fluctuations
Healthcare	patient recovery time
Machine learning	prediction variables
Quality control	number of defective items
Insurance	probability of accidents

Example in Data Science
Suppose a company predicts **daily website visitors**.
Random variable:
X = number of visitors per day
Possible values:
200, 350, 500, 1000
This is a **discrete random variable**.

11. Expected Value (Concept)
For a random variable, we often calculate the **expected value (mean)**.
It represents the **average outcome over many experiments**.
Example:
Expected number of heads in coin toss experiments.
This concept is widely used in:

- statistics
- risk analysis
- machine learning.

Final Concept Summary

Concept	Meaning
Random Variable	numerical representation of outcomes

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Discrete Random Variable	countable values
Continuous Random Variable	infinite values in a range

Concept Flow

Random Experiment
↓
Sample Space
↓
Random Variable
↓
Probability Distribution

Random variables allow us to **connect probability theory with real numerical data**.

Probability Density Function (PDF) & Cumulative Distribution Function (CDF)

After learning **random variables**, the next step is understanding **how probability is distributed across the values of a random variable**.

For this purpose, probability theory uses two important functions:

- **Probability Density Function (PDF)**
- **Cumulative Distribution Function (CDF)**

These functions are mainly used for **continuous random variables**.

1. Probability Density Function (PDF)

Definition: A **Probability Density Function (PDF)** describes how probability is distributed over the values of a **continuous random variable**.

It tells us **where values are more likely to occur**.

Notation

If random variable is **X**, the PDF is written as:

f(x)

Key Properties of PDF

Property	Explanation
PDF values are non-negative	$f(x) \geq 0$
Total area under the curve = 1	represents total probability
Probability between two values = area under curve between those values	used to calculate probabilities

Mathematical Property

$\int f(x)\, dx = 1$

This means the **total probability equals 1**.

2. Example of PDF

Suppose **X = height of students**.

The distribution might look like a **bell-shaped curve**.

Students with average height appear more frequently, while extremely short or tall heights are less common.

The **PDF curve shows this distribution of probability**.

Important Note

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For continuous random variables:
$P(X = x) = 0$
Because a continuous variable can take **infinitely many values**.
Instead we calculate probability over an **interval**.
Example:
$P(160 < X < 170)$

3. Cumulative Distribution Function (CDF)
Definition: The **Cumulative Distribution Function (CDF)** gives the probability that a random variable **X is less than or equal to a particular value**.

Notation : $F(x)$
Meaning: $F(x) = P(X \leq x)$

Interpretation
CDF tells us the **cumulative probability up to a certain value**.
Example: $F(10) = P(X \leq 10)$
This means probability that X is **less than or equal to 10**.

4. Relationship Between PDF and CDF
The PDF and CDF are mathematically related.
CDF from PDF
$F(x) = \int f(t)\, dt$
Meaning: CDF is the **integral of the PDF**.

PDF from CDF
$f(x) = d/dx\, [F(x)]$
Meaning: PDF is the **derivative of the CDF**.

5. Graphical Understanding
PDF
PDF shows the **density of probability at different values**.
Example shape:

```
        ^
     /   \
   /      \
  /        \
--|----------|--
```

Peak indicates where values are most common.

CDF
CDF shows **accumulated probability**.
Example shape:

```
0 ────┐
      |
      |
      └──── 1
```

It always:

- starts at **0**

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- increases continuously
- ends at **1**

6. Example

Suppose random variable **X = exam score**.

CDF might give:

Score	CDF Value
40	0.20
60	0.50
80	0.90

Interpretation:

- 20% students scored ≤ 40
- 50% students scored ≤ 60
- 90% students scored ≤ 80

7. Difference Between PDF and CDF

Feature	PDF	CDF
Full Form	Probability Density Function	Cumulative Distribution Function
Meaning	probability density at value	cumulative probability
Range	≥ 0	between 0 and 1
Graph shape	curve (bell-like)	increasing curve

8. Real-World Applications

Field	Application
Machine learning	modeling probability distributions
finance	stock return distributions
quality control	measurement errors
healthcare	biological measurements

Example in Data Science

Suppose:

X = time spent on a website

PDF shows:

- which time durations are most common.

CDF shows:

P(X ≤ 5 minutes)

meaning probability that a user leaves within **5 minutes**.

Final Summary

Concept	Meaning
PDF	probability density of continuous variable
CDF	cumulative probability up to value
Relationship	CDF = integral of PDF

Core Idea

Random Variable

↓

Probability Distribution

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↓
PDF → density of probability
CDF → cumulative probability

Binomial Distribution:

The **Binomial Distribution** is one of the most important **discrete probability distributions** in statistics and data science. It models situations where an experiment has **only two possible outcomes** repeated several times.
Examples of two outcomes:

- Success / Failure
- Yes / No
- Pass / Fail
- Head / Tail

1. Definition of Binomial Distribution
A **binomial distribution** describes the probability of obtaining a certain number of **successes in a fixed number of independent trials**, where each trial has the same probability of success.

2. Conditions for Binomial Distribution
A random experiment follows a binomial distribution if it satisfies these conditions.

Condition	Explanation
Fixed number of trials	experiment repeated **n times**
Two outcomes	success or failure
Constant probability	probability of success is same in every trial
Independent trials	result of one trial does not affect another

3. Examples of Binomial Experiments

Experiment	Success
Toss a coin	getting head
Student exam result	pass
Product quality check	defective item
Marketing campaign	customer buys product

Each trial results in **success or failure**.

4. Binomial Distribution Formula
The probability of getting **x successes in n trials** is:
p = probability of success , q =(1- p) failure

$$P(X = r) = nCr * p^r * (q)^{(n-r)}$$

Where:

Symbol	Meaning
n	number of trials
r	number of successes
p	probability of success
q	probability of failure
nCx	combination formula

5. Combination Formula

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The combination formula is:

$$nCr = n!/[r!(n-r)!]$$

Where **! (factorial)** means multiplication of numbers.

Example:

4! = 4 × 3 × 2 × 1

6. Example of Binomial Distribution

Example: Tossing a Coin 3 Times

Let:

Variable	Value
n	3
p	0.5
x	number of heads

We calculate probability of getting **2 heads**.

Step 1: Apply formula

$$P(X=2) = 3C2 \times (0.5)^2 \times (0.5)^1$$

Step 2: Calculate combination

3C2 = 3

Step 3: Final result

P(X=2) = 3 × 0.25 × 0.5

P(X=2) = 0.375

So probability of getting **2 heads in 3 coin tosses is 0.375**.

7. Binomial Distribution Table Example

Example: Coin tossed **3 times**

Number of Heads (x)	Probability
0	0.125
1	0.375
2	0.375
3	0.125

This table shows the **probability distribution of X**.

8. Mean and Variance of Binomial Distribution

Important formulas:

Measure	Formula
Mean	$\mu = np$
Variance	$\sigma^2 = npq$
Standard Deviation	$\sigma = \sqrt{np}$

Example

If:

n = 10

p = 0.5

Mean:

μ = np = 10 × 0.5 = 5

Variance:

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$\sigma^2 = npq$
$= 10 \times 0.5 \times 0.5$
$= 2.5$

9. Shape of Binomial Distribution
The shape depends on the probability **p**.

Value of p	Distribution Shape
p = 0.5	symmetrical
p < 0.5	positively skewed
p > 0.5	negatively skewed

For large values of **n**, binomial distribution approaches **normal distribution**.

10. Applications of Binomial Distribution

Field	Application
Quality control	defective products
Healthcare	disease testing
Marketing	customer response
Finance	credit approval
Machine learning	classification probability

Example in Data Science
Suppose:
X = number of customers who click an advertisement
Each customer either:

- clicks (success)
- does not click (failure)

The number of clicks follows a **binomial distribution**.

Final Summary

Concept	Meaning
Binomial Distribution	probability of successes in repeated trials
Trials	fixed number
Outcomes	success or failure
Mean	np
Variance	np(1-p)

Concept Flow
Repeated Experiments
↓
Two Possible Outcomes
↓
Count Number of Successes
↓
Binomial Distribution

Poisson Distribution

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The **Poisson Distribution** is a **discrete probability distribution** used to model the number of times an event occurs in a **fixed interval of time, space, or area**, when the events happen **independently and at a constant average rate**.
It is commonly used for **rare events**.

1. Definition : A **Poisson distribution** gives the probability of observing **x events in a fixed interval**, when the average number of occurrences is known.
Example situations:

- number of calls received by a call center in 1 minute
- number of accidents at a traffic signal in a day
- number of defects in a manufacturing process
- number of network failures per hour

2. Conditions for Poisson Distribution
Poisson distribution is used when these conditions are satisfied.

Condition	Explanation
Events occur randomly	no predictable pattern
Events are independent	one event does not affect another
Average rate is constant	events occur at constant mean rate
Rare events	probability of event in small interval is small

3. Poisson Distribution Formula
The probability of observing **r events** is:

$$P(X = r) = \left(e^{(-\lambda)} \times \lambda^r\right)/r!$$

Where:

Symbol	Meaning
X	number of occurrences
λ (lambda)	average number of events
e	exponential constant (≈ 2.718)
r!	factorial of r

4. Example
Suppose the **average number of phone calls received in a minute is 2**.
So: λ = 2
Find the probability of receiving **exactly 3 calls in a minute**.

Step 1: Apply formula

$$P(X = 3) = \left(e^{(-2)} \times 2^3\right)/3!$$

Step 2: Substitute values
$e^{-}2 \approx 0.1353$
3! = 6
P(X=3) = (0.1353 × 8) / 6

Step 3: Final result
P(X=3) ≈ 0.180
So the probability of receiving **3 calls in one minute is about 0.18**.

5. Mean and Variance of Poisson Distribution

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For Poisson distribution:

Measure	Formula
Mean	$\mu = \lambda$
Variance	$\sigma^2 = \lambda$
Standard Deviation	$\sigma = \sqrt{\lambda}$

Example

If: $\lambda = 4$

Then:

Measure	Value
Mean	4
Variance	4
Standard deviation	2

6. Poisson Distribution Table Example

Example: $\lambda = 2$

x (events)	Probability
0	0.135
1	0.270
2	0.270
3	0.180
4	0.090

This shows probabilities of different numbers of events.

7. Relationship with Binomial Distribution

Poisson distribution is often considered an **approximation of binomial distribution** when:

Condition	Requirement
n	very large
p	very small
$\lambda = np$	constant

This is called **Poisson approximation to binomial distribution**.

8. Graph Shape

The Poisson distribution graph is usually **right-skewed**, especially when λ is small.

When λ becomes larger, the distribution becomes more **symmetrical**.

9. Applications of Poisson Distribution

Field	Application
Telecommunications	number of phone calls per minute
Traffic analysis	number of accidents
Manufacturing	number of defective products
Healthcare	number of disease cases
Web systems	number of server requests

Example in Data Science

Suppose: X = number of website errors per hour

If the average number of errors is **$\lambda = 3$**, the Poisson distribution can predict the probability of **0, 1, 2, 3, or more errors**.

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This helps in **system reliability analysis**.

10. Key Differences: Binomial vs Poisson

Feature	Binomial Distribution	Poisson Distribution
Trials	fixed number	not fixed
Outcomes	success/failure	count of events
Parameter	n and p	λ
Application	repeated experiments	rare events in interval

Final Summary

Concept	Meaning
Poisson Distribution	probability of rare events
Parameter	λ (average event rate)
Mean	λ
Variance	λ

Concept Flow

Rare Events
↓
Average Rate (λ)
↓
Number of Occurrences
↓
Poisson Distribution

Geometric Distribution

The **Geometric Distribution** is a **discrete probability distribution** that models the number of trials required to obtain the **first success** in a sequence of independent Bernoulli trials.
A Bernoulli trial is an experiment with **two possible outcomes**:

- Success
- Failure

1. Definition: A **Geometric Distribution** gives the probability that the **first success occurs on the x-th trial**.
In simple terms:
Geometric distribution answers the question:
"How many trials are needed until the first success occurs?"

2. Conditions for Geometric Distribution

For an experiment to follow geometric distribution, the following conditions must be satisfied.

Condition	Explanation
Two outcomes	success or failure
Independent trials	one trial does not affect another
Constant probability	probability of success remains same
First success measurement	count trials until first success

3. Examples

Experiment	Success

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Toss a coin until first head appears	getting head
Customer purchase attempt	first successful purchase
Machine startup attempts	first successful start
Network transmission	first successful packet

4. Geometric Distribution Formula

The probability that the **first success occurs on the x-th trial** is:

P(X = x) =$q^{(x-1)} \times p$

Where:

Symbol	Meaning
X	number of trials until first success
p	probability of success
q	probability of failure
x	trial number

5. Example

Suppose a coin has probability of head:

p = 0.5

Find the probability that the **first head appears on the 3rd toss**.

Step 1: Apply formula

P(X = 3) = $(1 - 0.5)^{(3-1)} \times 0.5$

Step 2: Calculate

P(X = 3) = $(0.5)^2 \times 0.5$

P(X = 3) = 0.125

So the probability that the **first head appears on the third toss is 0.125**.

6. Probability Table Example

Example with:

p = 0.4

Trial (x)	Probability
1	0.40
2	0.24
3	0.144
4	0.086

As the number of trials increases, the probability decreases.

7. Mean and Variance of Geometric Distribution

Important formulas:

Measure	Formula
Mean	1 / p
Variance	$(1 - p) / p^2$
Standard deviation	$\sqrt{[(1 - p) / p^2]}$

Example

If:

p = 0.25

Mean: μ = 1 / 0.25 = 4

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This means **on average it takes 4 trials to get first success**.

8. Shape of Geometric Distribution
The geometric distribution is **right-skewed**.
Characteristics:

- highest probability occurs at **first trial**
- probability decreases as trials increase.

9. Applications of Geometric Distribution

Field	Application
Quality testing	number of trials before first defective item
Marketing	number of calls until first sale
Computer networks	packet transmission attempts
Reliability engineering	system startup attempts

Example in Data Science
Suppose a company sends marketing emails until a customer responds.
Random variable:
X = number of emails sent before first response
This follows a **geometric distribution**.

10. Difference Between Distributions

Distribution	Measures
Binomial Distribution	number of successes in fixed trials
Poisson Distribution	number of events in interval
Geometric Distribution	trials until first success

Final Summary

Concept	Meaning
Geometric Distribution	probability of first success
Parameter	p (probability of success)
Mean	1/p
Variance	$(1-p)/p^2$

Concept Flow
Repeated Trials
↓
Success or Failure
↓
Count Trials Until First Success
↓
Geometric Distribution

Uniform Distribution

The **Uniform Distribution** is a **continuous probability distribution** in which **all outcomes within a given interval are equally likely to occur**.
In other words, **every value in the interval has the same probability density**.

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1. Definition: A **Uniform Distribution** describes a situation where the probability of a random variable taking any value within a specific range is **constant**.
Example idea:
If a random variable lies between **a and b**, every value between them has **equal probability density**.

2. Example
Suppose: X = time at which a bus arrives
If the bus arrives **randomly between 10:00 and 10:10**, every time within that interval is equally likely.
So: 10:00 ≤ X ≤ 10:10
This follows a **uniform distribution**.

3. Probability Density Function (PDF)
For uniform distribution: $f(x) = 1 / (b - a)$
Where:

Symbol	Meaning
a	minimum value
b	maximum value
f(x)	probability density function

Conditions
$a \le x \le b$
Otherwise:
$f(x) = 0$

4. Example Calculation
Suppose:
$a = 2$
$b = 6$
Find the PDF value.
$f(x) = 1 / (6 - 2)$
$f(x) = 1 / 4$
So the probability density is **0.25**.

5. Probability Between Two Values
For uniform distribution:
$P(c \le X \le d) = (d - c) / (b - a)$
Where:

Symbol	Meaning
c	lower bound
d	upper bound

Example
Suppose:
$X \sim \text{Uniform}(0,10)$
Find:
$P(2 \le X \le 5)$
Solution:
$(5 - 2) / (10 - 0)$

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3 / 10 = 0.3
So probability = **0.3**.

6. Mean and Variance of Uniform Distribution
Important formulas:

Measure	Formula
Mean	$(a + b) / 2$
Variance	$(b - a)^2 / 12$
Standard Deviation	$\sqrt{[(b - a)^2 / 12]}$

Example
If:
$a = 2$
$b = 10$
Mean:
$\mu = (2 + 10) / 2 = 6$
Variance:
$\sigma^2 = (10 - 2)^2 / 12$
$\sigma^2 = 64 / 12 \approx 5.33$

7. Graph of Uniform Distribution
The PDF of uniform distribution is a **horizontal line**.

```
f(x)
  |
  | _________
  |
  |_________________
     a       b
```

Meaning:
- probability density is constant
- every value between **a and b** is equally likely.

8. Types of Uniform Distribution

Type	Description
Discrete Uniform Distribution	finite equally likely values
Continuous Uniform Distribution	values in continuous interval

Example:
Discrete uniform:
Rolling a fair dice
Continuous uniform:
Random arrival time

9. Applications of Uniform Distribution

Field	Example
Computer simulations	random number generation
Scheduling	random arrival times
Engineering	measurement uncertainty
Statistics	modeling equally likely events

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Example in Data Science
Random number generators used in algorithms often assume:
X ~ Uniform(0,1)
This distribution is used in:

- Monte Carlo simulations
- random sampling
- machine learning algorithms.

10. Key Characteristics

Feature	Description
Shape	rectangular
Probability density	constant
Range	between a and b
Symmetry	symmetrical distribution

Final Summary

Concept	Meaning
Uniform Distribution	equal probability across interval
PDF	$f(x) = 1/(b-a)$
Mean	$(a+b)/2$
Variance	$(b-a)^2/12$

Concept Flow
Random Variable
↓
Values in interval [a,b]
↓
Equal probability density
↓
Uniform Distribution

Parameters of Continuous Distributions

In probability and statistics, **parameters** describe the **characteristics of a probability distribution**.
For continuous distributions, parameters help define the **location, spread, and shape of the distribution**.
These parameters allow us to understand how the **values of a random variable behave**.

1. What are Parameters
Definition : Parameters are numerical values that describe the properties of a probability distribution.
They determine:

- where the distribution is centered
- how spread out the values are
- how the distribution is shaped

2. Common Parameters of Continuous Distributions
The most important parameters are:

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Parameter	Symbol	Meaning
Mean	μ (mu)	average value of the distribution
Variance	σ^2	spread of values around the mean
Standard Deviation	σ	square root of variance

These parameters describe **central tendency and dispersion** of a continuous distribution.

3. Mean of Continuous Distribution

Definition: The **mean** is the expected or average value of a continuous random variable.
It represents the **center of the distribution**.

Formula

For continuous random variable:

$\mu = \int x\, f(x)\, dx$

Where:

Symbol	Meaning
x	random variable
f(x)	probability density function
μ	mean

Interpretation: The mean indicates the **central location of the probability distribution**.
Example: If the mean height of students is **170 cm**, most students' heights are around this value.

4. Variance of Continuous Distribution

Definition: Variance measures **how much the values deviate from the mean**.
It describes the **spread of the distribution**.

Formula

$\sigma^2 = \int (x - \mu)^2 f(x)\, dx$

Where:

Symbol	Meaning
σ^2	variance
μ	mean
f(x)	probability density function

Interpretation

Variance Value	Meaning
Small variance	data values close to mean
Large variance	data widely spread

5. Standard Deviation

Definition : The **standard deviation** is the square root of variance.
It measures the **average distance of data values from the mean**.

Formula $\sigma = \sqrt{\sigma^2}$

Interpretation

Standard deviation helps understand **how concentrated the values are around the mean**.
Example:
If the standard deviation is small, the values are **closely clustered around the average**.

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6. Example

Suppose a continuous random variable represents **students' heights**.

Parameter	Value
Mean	170 cm
Variance	25
Standard deviation	5 cm

Interpretation:

Most heights fall within **165–175 cm**.

7. Role of Parameters in Continuous Distributions

Parameters define the **shape and behavior of distributions**.

Examples:

Distribution	Parameters
Uniform Distribution	a, b
Normal Distribution	μ, σ
Exponential Distribution	λ

Changing parameter values changes the **distribution shape**.

8. Example: Uniform Distribution Parameters

For uniform distribution:

$X \sim U(a, b)$

Parameters:

Parameter	Meaning
a	minimum value
b	maximum value

These parameters define the **range of the distribution**.

9. Importance in Data Science

Parameters are important because they help:

Application	Purpose
describe distributions	summarize data behavior
model real-world phenomena	represent probability patterns
build statistical models	estimate unknown values
train machine learning models	understand data variability

Many machine learning models rely on **parameter estimation**.

Final Summary

Parameter	Purpose
Mean (μ)	center of distribution
Variance (σ^2)	spread of data
Standard Deviation (σ)	variability measure

Concept Flow

Continuous Random Variable

↓

Probability Distribution

↓

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Parameters
↓
Mean + Variance + Standard Deviation
These parameters describe **how the probability distribution behaves**.

Exponential Distribution

The **Exponential Distribution** is a **continuous probability distribution** used to model the **time between events** in a process where events occur **randomly and independently at a constant average rate**.
It is commonly used for modeling **waiting times**.

1. Definition: The **Exponential Distribution** describes the probability of the **time until the next event occurs**.
Examples of such events:

- time between customer arrivals in a queue
- time between system failures
- time between phone calls at a call center
- time between website requests

2. Parameter of Exponential Distribution
The exponential distribution has **one parameter**.

Parameter	Symbol	Meaning
Rate parameter	λ (lambda)	average number of events per unit time

Example:
λ = 3
This means **3 events occur per unit time on average**.

3. Probability Density Function (PDF)
The PDF of exponential distribution is:
f(x) = $\lambda e^{(-\lambda x)}$
Where:

Symbol	Meaning
x	time until event occurs
λ	rate parameter
e	exponential constant (≈ 2.718)

Condition:
x ≥ 0

4. Cumulative Distribution Function (CDF)
The CDF gives the probability that the event occurs **before or at time x**.
F(x) = $1 - e^{(-\lambda x)}$
Meaning:
P(X ≤ x)
Probability that the event occurs **within time x**.

5. Mean and Variance
Important formulas for exponential distribution:

Measure	Formula

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Mean	$1 / \lambda$
Variance	$1 / \lambda^2$
Standard deviation	$1 / \lambda$

Example
Suppose: $\lambda = 2$
Mean waiting time:
Mean = 1 / 2 = 0.5
Variance: Variance = 1 / 4 = 0.25

6. Graph Shape
The exponential distribution graph is **right-skewed**.

```
f(x)
 |
 |\
 | \
 |  \
 |   \
 |    \
 | ____\____________
      x
```

Characteristics:

- highest probability near **x = 0**
- probability decreases as **x increases**

7. Memoryless Property
One special feature of exponential distribution is the **memoryless property**.
This means:
$P(X > s + t \mid X > s) = P(X > t)$
Interpretation:
The probability of waiting **t more units of time** does not depend on how long we have already waited.
Example:
If a machine has not failed for 5 hours, the probability of failure in the next hour **remains the same**.

8. Relationship with Poisson Distribution
Exponential distribution is closely related to **Poisson distribution**.

Distribution	What it Measures
Poisson Distribution	number of events in an interval
Exponential Distribution	time between events

If events follow a **Poisson process**, the time between events follows an **exponential distribution**.

9. Applications of Exponential Distribution

Field	Application
Queue theory	waiting time for service
Reliability engineering	time until system failure
Telecommunications	time between packet arrivals
Web analytics	time between user actions

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Data science	event arrival modeling

Example in Data Science
Suppose:
X = time between website requests
If requests arrive at an average rate:
λ = 5 requests per minute
Then the waiting time between requests follows an **exponential distribution**.

10. Comparison with Other Distributions

Distribution	Measures
Binomial Distribution	number of successes in fixed trials
Poisson Distribution	number of events in interval
Geometric Distribution	trials until first success
Exponential Distribution	time until next event

Final Summary

Concept	Meaning
Exponential Distribution	waiting time between events
Parameter	λ (rate of events)
Mean	1 / λ
Variance	1 / λ^2
PDF	$\lambda e^{(-\lambda x)}$

Concept Flow
Poisson Process
↓
Events Occur Randomly
↓
Time Between Events
↓
Exponential Distribution

Chi-Square Distribution

The **Chi-Square (χ^2) Distribution** is a **continuous probability distribution** widely used in **statistical inference and hypothesis testing**.
It is mainly used to analyze **categorical data and variance**.
It plays an important role in tests such as:

- **Goodness-of-fit test**
- **Test of independence**
- **Test of variance**

1. Definition: The **Chi-Square distribution** is the distribution of the **sum of squares of independent standard normal random variables**.
In simple terms:
If we square several independent normal variables and add them together, the result follows a **Chi-Square distribution**.

2. Symbol

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The Chi-Square distribution is represented by: χ^2

3. Parameter of Chi-Square Distribution

The Chi-Square distribution has **one parameter**.

Parameter	Meaning
Degrees of Freedom (df)	number of independent values in the data

4. Degrees of Freedom

Definition: Degrees of freedom (df) represent the number of values that are free to vary in the calculation.

Example: If we know the sum of 3 numbers is **30**, and two numbers are **10 and 8**, the third number must be **12**.

So only **2 values are free to vary**.

Therefore:

df = n − 1

5. Shape of Chi-Square Distribution

The shape of the Chi-Square distribution depends on **degrees of freedom**.

Characteristics:

Feature	Description
Range	values ≥ 0
Shape	right-skewed
Symmetry	not symmetrical
As df increases	distribution becomes more symmetrical

Graph Shape

```
f(x)
 |
 |    /\
 |   /  \
 | /     \
 |/       \____
 |____________
      x
```

The distribution starts at **0** and extends to positive values.

6. Mean and Variance

For Chi-Square distribution:

Measure	Formula
Mean	df
Variance	2 × df
Standard deviation	√(2 × df)

Example

If:

df = 4

Mean:

μ = 4

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Variance:
$\sigma^2 = 2 \times 4 = 8$
Standard deviation:
$\sigma = \sqrt{8} \approx 2.83$

7. Applications of Chi-Square Distribution
Chi-Square distribution is used in several statistical tests.

Test	Purpose
Chi-Square Goodness-of-Fit Test	check if data follows expected distribution
Chi-Square Test of Independence	check relationship between categorical variables
Chi-Square Test of Homogeneity	compare distributions of different populations
Variance testing	evaluate population variance

8. Example: Chi-Square Test of Independence
Suppose a survey studies the relationship between **gender and product preference**.

Gender	Product A	Product B
Male	40	20
Female	30	10

Chi-Square test helps determine whether:
Gender and product preference are independent
or related.

9. Importance in Data Science
Chi-Square distribution is useful in:

Field	Application
Machine learning	feature selection
statistics	hypothesis testing
data analysis	categorical variable relationships
bioinformatics	gene association studies

Example:
In **feature selection**, the Chi-Square test helps identify important variables for classification models.

10. Key Characteristics

Feature	Description
Distribution type	continuous
Range	≥ 0
Shape	right-skewed
Parameter	degrees of freedom

Final Summary

Concept	Meaning
Chi-Square Distribution	distribution of squared normal variables
Parameter	degrees of freedom
Mean	df
Variance	2df

Concept Flow

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Normal Random Variables
↓
Square the Variables
↓
Sum of Squares
↓
Chi-Square Distribution

Student's t-Distribution

The **Student's t-Distribution** is a **continuous probability distribution** used when we want to estimate the **population mean** from a **small sample size** and the **population standard deviation is unknown**.
It is commonly used in **hypothesis testing and confidence intervals**.

1. Definition: The **Student's t-Distribution** describes the distribution of the **sample mean when the sample size is small and the population variance is unknown**.
It is similar to the **normal distribution**, but it has **heavier tails**.

2. Origin of the Name
The distribution was introduced by statistician **William Sealy Gosset**.
He published his work under the pseudonym **"Student"**, which is why it is called the **Student's t-Distribution**.

3. When to Use t-Distribution
The t-distribution is used when the following conditions are met.

Condition	Explanation
Sample size is small	usually $n < 30$
Population variance unknown	standard deviation not known
Data approximately normal	data follows normal distribution

4. Formula of t-Statistic
The t-value is calculated using:
$t = (\bar{x} - \mu) / (s / \sqrt{n})$
Where:

Symbol	Meaning
$\bar{x}$	sample mean
μ	population mean
s	sample standard deviation
n	sample size

5. Degrees of Freedom
The Student's t-distribution depends on **degrees of freedom (df)**.
$df = n - 1$
Where **n** is the sample size.

Example
If sample size: $n = 10$
Degrees of freedom: $df = 9$

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6. Shape of Student's t-Distribution

The t-distribution is **similar to the normal distribution**, but with **heavier tails**.
This means it accounts for **greater uncertainty when sample size is small**.
Characteristics:

Feature	Description
Symmetry	symmetric around mean
Shape	bell-shaped
Tails	heavier than normal distribution
Mean	0

Graph Concept

```
      ^
     / \
    /   \
   /     \
---|---------|---
```

As **sample size increases**, the t-distribution approaches the **normal distribution**.

7. Mean and Variance

For Student's t-distribution:

Measure	Value
Mean	0
Variance	df / (df − 2) (df > 2)

8. Applications of Student's t-Distribution

Application	Purpose
t-test	compare sample mean with population mean
confidence intervals	estimate population mean
small sample analysis	statistical inference
machine learning evaluation	parameter estimation

9. Types of t-Tests

Student's t-distribution is used in three main tests.

Test	Purpose
One-sample t-test	compare sample mean with population mean
Independent t-test	compare two different groups
Paired t-test	compare measurements before and after experiment

10. Example

Suppose:

Parameter	Value
Sample mean	50
Population mean	45
Standard deviation	10
Sample size	25

Calculate t-value.
Step 1: $\sqrt{n} = \sqrt{25} = 5$
Step 2: $t = (50 - 45) / (10 / 5)$

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Step 3: t = 5 / 2 = 2.5
So the **t-statistic = 2.5**.

11. Difference Between Normal and t-Distribution

Feature	Normal Distribution	t-Distribution
Sample size	large	small
Population variance	known	unknown
Shape	narrower	heavier tails
Parameter	mean & variance	degrees of freedom

12. Importance in Data Science
Student's t-distribution is widely used in:

Field	Application
data science	hypothesis testing
machine learning	model evaluation
research	small sample analysis
statistics	confidence intervals

Final Summary

Concept	Meaning
Student's t-Distribution	distribution used for small samples
Parameter	degrees of freedom
Mean	0
Shape	bell-shaped with heavy tails

Concept Flow
Small Sample Data
↓
Unknown Population Variance
↓
Estimate Mean Using Sample
↓
Student's t-Distribution

F-Distribution:

The **F-Distribution** is a **continuous probability distribution** used mainly to **compare variances of two populations**.
It plays an important role in **statistical hypothesis testing**, especially in **ANOVA (Analysis of Variance)**.

1. Definition: The **F-distribution** is the distribution of the **ratio of two independent chi-square variables divided by their respective degrees of freedom**.
In simpler terms: It measures how **two variances compare with each other**.

2. Formula of F-Statistic
The F-statistic is calculated as:
F = (Variance of sample 1) / (Variance of sample 2)
More formally:

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$F = (s_1^2 / df_1) / (s_2^2 / df_2)$

Where:

Symbol	Meaning
s_1^2	variance of first sample
s_2^2	variance of second sample
df_1	degrees of freedom of sample 1
df_2	degrees of freedom of sample 2

3. Parameters of F-Distribution

The F-distribution depends on **two parameters**.

Parameter	Meaning
df_1	numerator degrees of freedom
df_2	denominator degrees of freedom

These parameters determine the **shape of the distribution**.

4. Characteristics of F-Distribution

Feature	Description
Distribution type	continuous
Range	values ≥ 0
Shape	right-skewed
Parameters	two degrees of freedom

The distribution **cannot take negative values** because it is based on **variance ratios**.

5. Graph Shape

The F-distribution is **positively skewed**.

```
f(x)
 |
 |     /\
 |    /  \
 |   /    \
 |  /      \____
 |_____________
      x
```

As the **degrees of freedom increase**, the curve becomes **more symmetrical**.

6. Mean of F-Distribution

The mean exists when:

$df_2 > 2$

Formula:

$\text{Mean} = df_2 / (df_2 - 2)$

7. Applications of F-Distribution

The F-distribution is widely used in statistical tests.

Application	Purpose
F-test	compare variances of two populations
ANOVA	compare means of multiple groups
Regression analysis	test overall model significance
Experimental design	analyze treatment effects

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8. Example: Comparing Variances

Suppose two machines produce items.

Machine	Variance
Machine A	16
Machine B	9

F-statistic:

F = 16 / 9

F ≈ 1.78

This value is compared with an **F-distribution table** to determine whether the variances differ significantly.

9. Example in Data Science

In **regression analysis**, the F-test checks whether the **overall model is statistically significant**.

Example:

F-statistic = variance explained by model / variance of error

If the F-value is large, it means the **model explains the data well**.

10. Relationship with Other Distributions

The F-distribution is closely related to other probability distributions.

Distribution	Relationship
Chi-Square Distribution	F-distribution is ratio of two chi-square variables
t-Distribution	square of t-distribution follows F-distribution
Normal Distribution	both chi-square and t derive from normal distribution

11. Difference Between t-Distribution and F-Distribution

Feature	t-Distribution	F-Distribution
Purpose	test mean differences	test variance differences
Parameters	one degree of freedom	two degrees of freedom
Shape	symmetric	right-skewed

Final Summary

Concept	Meaning
F-Distribution	ratio of two variances
Parameters	df_1 and df_2
Range	≥ 0
Application	ANOVA and variance comparison

Concept Flow

Sample Variances

↓

Ratio of Variances

↓

F-Statistic

↓

F-Distribution

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